The Economics of Composable Options Markets

Replication, liquidity formation, and capital efficiency in composable financial markets

World Markets Research

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Abstract

Onchain options are primarily constrained by liquidity. It is possible to construct options using existing onchain spot and loan markets. October 2026 market data suggests a 6-12x increase in top of book liquidity is possible for onchain calls and puts, relative to Derive V2, by constructing options from this existing liquidity. This opportunity would allow traders to exploit market inefficiencies in current onchain options markets. We explore how the unique properties of fully onchain matching and settlement unlock a more efficient market structure intermediated by dealers in traditional equities and FX options markets.

1 Introduction

European options are not economically isolated contracts. Their payoffs are linked to spot, fixed-term financing, forwards, and other options via no-arbitrage identities. These relationships permit economically equivalent payoff profiles to be constructed through different instruments; the only difference between the "constructed" payoff and "an option" is label, which is trivially solved. The amount of liquidity available for a particular option therefore need not be determined solely by market makers' resting orders in that option's book.

The opportunity is to integrate replication principles into a trading venue, rather than rely only on market makers. A venue can compare a native option quote against an executable combination of spot, financing, and an opposite option, construct the cheaper or deeper route, and deliver the requested payoff under enforceable settlement conditions. An onchain exchange is directly able to source existing liquidity of the underlying spot and loan markets adherent to replication logic. Where exact static construction is unavailable, a liquidity provider can supply the exposure dynamically through the same spot and perpetual markets. Onchain portfolio margin for spot, loans, and options minimizes the additional capital required to supply the exposures which cannot be statically replicated.

This paper develops that architecture from Stoll's (1969) put-call relationships, Black and Scholes' (1973) and Merton's (1973) dynamic replication, and Merton's (1974) debt-option equivalence [1-4]. It then uses current onchain spot, lending, and options data to quantify plausible quote capacity and execution benefits relative to current order book depth on Derive. Derive is a demanding reference: it already offers options and perpetual portfolio margin, cross-asset collateral, and RFQ execution [11,12]. The analysis therefore does not attribute those existing capabilities to a new design.

This paper evaluates the potential gains in executable liquidity and capital efficiency under the assumption that the necessary counterparties, constituent-market liquidity, financing, and risk capital are available at economically viable prices. Its objective is to quantify the improvements achievable through replication and integrated market structure under these conditions.

Three questions organize the paper: Which option payoffs can a venue construct exactly from executable constituent markets? How do dynamic hedging and joint margin extend liquidity beyond static replication? And, under explicitly specified inputs, how much additional option depth and immediate execution could these mechanisms support?

2 Replication Across Financial Markets

2.1 Put-call parity and ordinary financing

For matched European calls and puts on the same underlying, strike K, and expiry T, the terminal payoff identity is

$$C_T - P_T = S_T - K. \tag{1}$$

Let $B_t(K,T)$ be the present value of an enforceable fixed cash payment of K at T. Assuming matching settlement, no intervening cash distributions, and financeable positions, the corresponding price relation is

$$C_t - P_t = S_t - B_t(K,T). \tag{2}$$

The right-hand side is the financed spot, or forward, component of option payoffs. A conventional recourse loan can fund the spot purchase and supply the fixed repayment obligation. Rearranging gives several exact

Table 1. Matched-payoff constructions at expiry

Target payoff	Equivalent portfolio
Long call	Long spot + long put - fixed repayment claim
Long put	Long call - spot + fixed repayment claim
Long forward	Long call - long put = spot - fixed repayment claim
Covered call	Long spot + short call = fixed repayment claim + short put

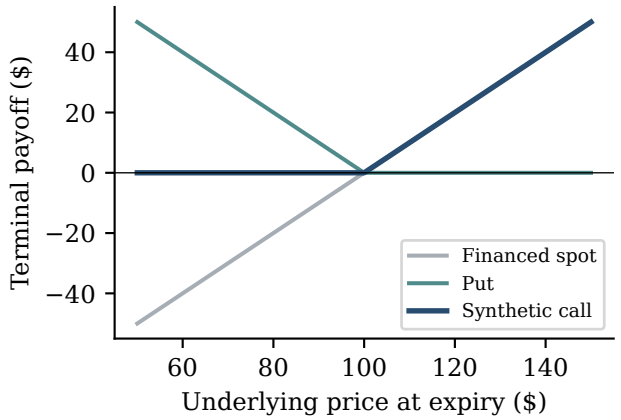


Figure 1: Exact payoff construction at strike $K = 100$: financed spot plus a put equals a call. Illustrative terminal payoffs; premiums excluded.

static constructions, given a blockchain gives all legs identical settlement properties.

Here a minus sign denotes a short position or a repayment obligation, not the absence of collateral. These identities relate contractual payoffs, not necessarily the margin requirements or bid-ask prices of the corresponding trades, this paper explores margin requirements in a subsequent section. Exact static option construction needs a source of convexity: the opposite option, or another enforceable claim with the same contingent payoff [1,23].

2.2 Non-recourse debt as an alternative contingent claim

Merton's (1974) debt-option relationship identifies one such alternative. For an idealized, maturity-settled **non-recourse** loan secured by an asset worth S_T and promising K at T, the lender receives

$$D_T = min\left(S_T, K\right) = K - max\left(K - S_T, 0\right). \quad (3)$$

The lender's claim equals a fixed repayment **minus a put**; the borrower's retained equity is $\max\left(S_T - K, 0\right)$, a call. For example, a \$90 repayment secured by collateral ending at \$120 or \$70 yields lender receipts of \$90 or \$70, and borrower equity of \$30 or \$0. This is an exact contingent-payoff relationship, not a statement that a conventional liquidatable DeFi loan embeds a freely tradable European put. The example is relevant as an illustration of the broader body of work around option replication.

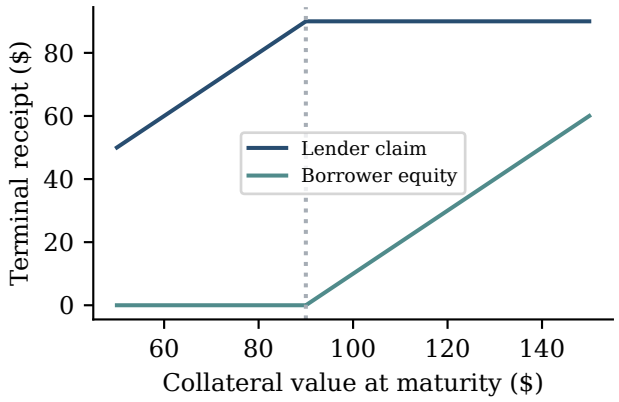


Figure 2: Non-recourse debt with promised repayment of \$90 splits collateral into a lender claim and borrower equity. Illustrative maturity-settled contract.

2.3 Dynamic replication where no exact static route exists

When neither an opposite option nor a matching contingent claim is executable, Black-Scholes-Merton replication provides another route: a liquidity provider assumes an options position and dynamically adjusts its underlying exposure [2,3]. For option value $V(S,t)$, the instantaneous directional hedge is determined by

$$\Delta_t = \frac{\partial V}{\partial S}. \quad (4)$$

A dealer short an approximately at-the-money call with delta 0.5 initially acquires roughly half a unit of spot or equivalent long-perpetual exposure per call. That hedge must change as price, time, and implied volatility evolve. A spot-plus-financing hedge corresponds to the idealized replication model; a perpetual introduces funding and basis risks. A practical short-option dealer still bears discontinuous rehedging, gamma, volatility repricing, execution, and liquidation risks.

Thus static construction transfers an already available equivalent payoff; dynamic replication creates an additional quoted option position whose residual risks must be borne and financed.

3 Constructing Synthetic Options Liquidity

The exchange-level proposition is to "construct" resting limit orders in the native options book as a function of all executable, payoff-equivalent routes, then atomically execute the best feasible route or combination. Onchain, programmatic settlement creates the enforceable conditions under which the synthetic route is not only valid but economically indistinguishable from the "native" option contract.

3.1 From parity to executable bids and asks

Consider a call and put with matching strike and expiry. Let S^a, S^b and P^a, P^b denote **executable** spot and put asks and bids. Let B^{borrow} be the cash actually received today for assuming a fixed repayment of K at expiry, and B^{lend} the cash paid today to acquire the corresponding fixed-payment claim. With financing and credit frictions, generally $B^{borrow} \leq B^{lend}$. The parity-implied executable call prices are

$$C^{syn,a} = S^a + P^a - B^{borrow}, \tag{5}$$

$$C^{syn,b} = S^b + P^b - B^{lend}. \tag{6}$$

The first builds a **long** synthetic call by buying spot and a put while assuming the repayment obligation. The second builds a **short** synthetic call by selling the corresponding constituent positions and acquiring the fixed claim; it requires inventory or executable shorting and appropriate short-put margin. Both directions include trading fees, slippage, and settlement costs, which are all deterministically knowable at runtime.

For illustration, take spot bid/ask \$99.80/\$100.20, put bid/ask \$3.60/\$4.00, financing proceeds \$99.80, and the cost of the fixed-payment claim \$99.90, all for a \$100 strike and one underlying unit. Synthetic call bid/ask is therefore **\$3.50/\$4.40**. Against a native call bid/ask of \$3.25/\$4.80, the synthetic route improves both sides. This is arithmetic from assumed *firm* constituent quotes, not an observed onchain arbitrage.

3.2 Depth, aggregation, and execution

Executable depth is constrained by the scarcest constituent leg. If an ask-side call route has 150 units available at the spot ask, 80 at the put ask, and sufficient term financing for only 60, then its synthetic call capacity is **60 units**, not 150. More generally, in contract-equivalent units,

$$q_{syn} \leq \min\left(q_{spot}, q_{opposite\ option}, q_{financing}\right). \tag{7}$$

If the native call ask also offers 40 independent units, the venue can execute up to 100 units **within a limit price that admits both routes**. At a tighter limit only the synthetic units may qualify. Routing can atomically account for price impact, shared constituent inventory, credit constraints, and whether other routes already consume the same spot or financing liquidity; otherwise quoted depth is double-counted.

3.3 Adding dynamically hedged quotations

Exact synthetic depth is limited by the supply of existing convexity. Where that supply is absent or too expensive, a risk-bearing dealer can offer further options capacity against dynamically executed spot or perpetual hedges. The venue may help price and coordinate this route, but **some party must own the residual short- or long-volatility risk**. Dynamic hedging is not an instantaneously enforceable parity trade.

The venue therefore aggregates two different liquidity sources: **payoff-equivalent static routes**, which repackage existing positions, and **risk-backed dealer quotations**, which expand options risk inventory. Shared hedge markets and a common margin account make the second source more capital-efficient.

4 Dynamic Hedging and Portfolio Margin

Options fragment into strikes, maturities, and call/put types while their directional hedges concentrate in the same underlying markets. A menu of 20 strikes, 10 expiries, and two types comprises 400 option series, yet an options portfolio has one aggregate underlying delta per risk asset:

$$q_{hedge} \simeq -n_j \Delta_j. \tag{8}$$

The dealer can source that delta from spot or perpetual liquidity rather than separately hold a directional hedge for every series. Its remaining exposure includes gamma, volatility-surface changes, basis, funding, and execution. A common margin engine can measure these together with the spot and financing positions, rather than simply sum independent position margins.

4.1 Net risk and eligible collateral are distinct

Netting reduces the margin associated with offsetting exposures. Collateral recognition determines which existing assets may satisfy that margin. In an illustrative stress framework,

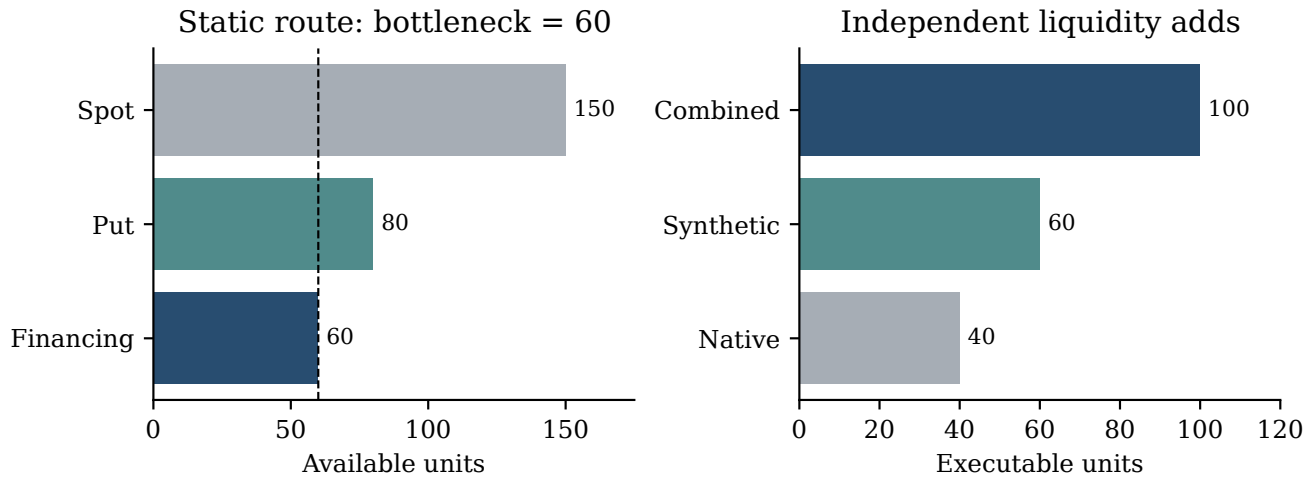


Figure 3: The static route is limited by its smallest constituent. Independent native liquidity can add to synthetic depth at an admissible price limit. Section 3.2 assumptions.

$$M(\Pi) = \max_{s \in S} \max\left(0, -\Delta V(\Pi; s)\right), \tag{9}$$

where Π is the entire portfolio and S is a common set of price, volatility, funding, liquidity, and credit scenarios. An integrated account can recognize a covered call's underlying, a protective put's downside protection, or the offset between spot and a short perpetual. It must also preserve solvency during adverse intermediate price paths.

Derive already recognizes options and perpetual netting and certain spot assets as collateral [11]. A potential incremental mechanism is to recognize eligible **native fixed-term lender claims** in the same ledger as spot, perps, and options, subject to their liquidity and credit risk. The illustrative integrated-credit scenario uses 98% recognition of an eligible lender claim, a documented existing multi-asset collateral parameter [20-22]; an options extension still requires joint stress treatment.

4.2 A matched financing example

Suppose a dealer has a $1.00 million eligible stable-denominated fixed-term lending claim, sells $1.00 million underlying-reference notional of at-the-money calls, and holds a $0.50 million long-perpetual hedge. For the simplified expiry prices $70 and $130 from an initial $100, the derivative book loses $150,000 in either shock. Assign **the same** $150,000 illustrative risk requirement to both compared derivatives accounts.

The illustrative **13.04%** total-capital reduction is $0.15M/$1.15M. It comes from reusing an existing credit asset, **not** from lowering the derivatives loss estimate. With a fixed total capital budget of $1.45 million, of which $1 million is placed in term credit, a further simplified 15% derivatives margin factor yields capacities of $0.45M/0.15=**$3.00M** and ($0.45M+$0.98M)/0.15=**$9.53M**, or **3.18x**. These are model portfolios contrasting eligible and ineligible fixed-term claims; they are not measured Derive V3 or proposed V2 margin outputs.

Collateral reuse adds liquidity and wrong-way risks. Joint stress must mark down the lending claim when the derivative portfolio is impaired, incorporate settlement and liquidation costs, and avoid counting the same asset twice.

4.3 From hedges and capital to dealer quote size

For option delta magnitude $|\Delta|$, executable underlying hedge notional H, eligible risk-financing capital C_e, approximate margin fraction m, and independent residual-risk ceiling N_R, a useful **dynamic dealer capacity bound** is

$$Q_{dyn} \leq \min\left(\frac{H}{|\Delta|}, \frac{C_e}{m}, N_R\right). \tag{10}$$

This is not a formula for static synthetic depth, which depends on each constituent order book and financing route. Portfolio margin relaxes the capital constraint.

5 Quantifying the Liquidity Advantage

The empirical benchmark is Derive, an established on-chain options venue with portfolio margin and RFQ [11-13]. We use an **October 12, 2026 research snapshot** to retain a consistent numerical comparison. It is a dated set of market observations. We distinguish observed volume and positions from executable depth, assumed liquidity-provider capacity, and modeled customer

Table 2. Incremental collateral recognition for an identical dealer portfolio

Measure	Fixed-term claim external to derivatives account	Integrated eligible term claim
Existing term-lending investment	$1.00M	$1.00M
Netted derivatives requirement	$0.15M	$0.15M
Term claim eligible against requirement	$0	$0.98M
Additional cash needed	$0.15M	$0
Total capital committed	$1.15M	$1.00M

fills.

5.1 Onchain markets and the Derive benchmark

Data boundary: The October 12 values are dated figures transcribed from the cited source dashboards; without an immutable historical export, they cannot be reconstructed from the current live counters. The Sept. 24 source captures **public visible order-book depth**, not Derive V3 or firm institutional RFQ liquidity. Trading volume is a time-period **flow**; outstanding loans and open interest are point-in-time **stocks**. Neither is a measure of immediately executable hedge depth.

The corresponding ratios - spot turnover to options turnover, combined active loans to Derive option open interest, and Derive's share of tracked onchain option turnover - are

$$\frac{246.593}{4.881} = 50.5. \tag{11}$$

$$12.15 + 5.78 = 17.93, \qquad \frac{17.93}{1.655} = 10.8. \tag{12}$$

$$\frac{4.236}{4.881} = 86.8\%. \tag{13}$$

These ratios establish the **relative scale** of existing spot, credit, and option markets. They do not imply 50.5 times more executable hedge liquidity, or 10.8 times more reusable option collateral. Only price-and-size executable orders and eligible, maturity-matched funding can enter an actual synthetic route.

5.2 The static-route mechanism and what the data establish

Section 3 gives executable synthetic call prices and a constituent-depth bound. To measure the **static-routing uplift** for a given contract, one would observe spot and opposite-option order-book levels, firm ten-day lending and repayment prices, fees, and permitted short positions **at the same instant**. Native and synthetic routes could then be sorted by all-in price, allocated without reusing the same constituent liquidity, and compared at an identical client price limit.

The aggregate spot and loan data in Table 3 establish an economically material **candidate liquidity base**, but do not provide those simultaneous per-strike inputs. The executable parity example in Section 3 demonstrates the mechanism and its pricing, not its observed market size.

5.3 Potential liquidity improvement through jointly margined option portfolios

A portfolio-based calibration identifies potential quoted quantities of 532–1,052 option units, approximately 6–12 times the 90-unit ask in the historical Derive V2 benchmark. These quantities use approximately the same 60 ETH of initial spot hedging, with measured spot execution costs equivalent to approximately $0.30 and $0.15 per customer-facing option respectively. In an illustrative stress model, the associated risk reserves are approximately $19,000, supporting 1.34 - 2.64 million of customer-facing option reference notional. A more aggressive scenario reaches approximately 2,620 units, or 29 times the historical displayed size, but requires substantially more liquidity in a closely matching hedge option.

These estimates describe conditional quotation capacity. They combine measured onchain spot execution with modeled option portfolios and assume that sufficient offsetting options and financing are executable. The 6–12-fold range represents two specified hedge configurations rather than a statistical confidence interval. Gas costs are set to zero throughout.

The mechanism: finance the residual portfolio risk

The larger opportunity arises when a dealer combines the customer-facing option with an offsetting option and hedges only the remaining directional exposure in spot. For example, a dealer offering an at-the-money call can simultaneously acquire a call with the same expiry and a slightly higher strike. The customer receives the requested call, while the dealer holds a short call spread and a spot hedge.

The purchased call offsets much of the sold call's

Table 3. Market observations used as inputs

Market	Measurement	Value	Period / date	Source
Onchain spot DEXs	Trading notional	$246.593B	Prior 30 days; Oct. 12, 2026	[15]
Aave V3	Active loans outstanding	$12.15B	Snapshot; Oct. 12, 2026	[18]
Morpho	Active loans outstanding	$5.78B	Snapshot; Oct. 12, 2026	[17]
All tracked onchain options	Option notional traded	$4.881B	Prior 30 days; Oct. 12, 2026	[14]
Derive	Option notional traded	$4.236B	Prior 30 days; Oct. 12, 2026	[13]
Derive V3	Options open interest	$1.655B	Snapshot; Oct. 12, 2026	[13]
Derive V2	Best bid on ETH-20261002-2700-C	82.8 ETH	Public capture; Sep. 24, 2026	[16]
ETH option normalization	Contract strike	$2,700	Same historical option	[16]

gamma and vega. The option spread alone also has a bounded terminal liability determined by the difference between the strikes. Adding the spot hedge reduces initial delta but changes the portfolio's risk profile, so the combined position must be assessed jointly. [24]

A unified margin framework could recognize these offsets and finance the resulting portfolio according to its residual risk. This extends the existing principle of evaluating spot, perpetuals, and borrowing together. [20] The economic benefit is that capital need not be reserved as though each leg were an independent directional exposure.

For q customer-facing calls, the initial spot purchase is:

$$h = q\,(\Delta_{\text{sold}} - \Delta_{\text{hedge}})\,. \tag{14}$$

A standalone short call with delta near 0.50 requires approximately half an ETH of initial hedging per option unit. When a nearby call supplies an offsetting position, the difference between the two deltas can be much smaller. The same executable spot liquidity can therefore support a substantially larger customer-facing quotation.

Calibration using onchain spot liquidity

The spot input is a simulated purchase of 60 ETH across two Uniswap V3 WETH/USDC pools at Ethereum block 26,189,586, corresponding to October 14, 2026, at 08:36:59 UTC. The route buys 40 ETH through the 5-basis-point pool and 20 ETH through the 1-basis-point pool. At the common reference price of 2,510.8339 USDC per ETH, the quoted cash input is approximately $150,808.54, including $158.51 of pool fees and price impact. The calls are evaluated against the same blockchain state. [27,28]

To translate this observation into option capacity, consider hypothetical seven-day European calls with an at-the-money customer strike, 50% implied volatility, and a flat volatility surface. Black–Scholes valuation uses zero interest rates; actual borrowing expense is introduced separately. Each customer-facing call is backed by one purchased call with the same expiry and a higher strike.

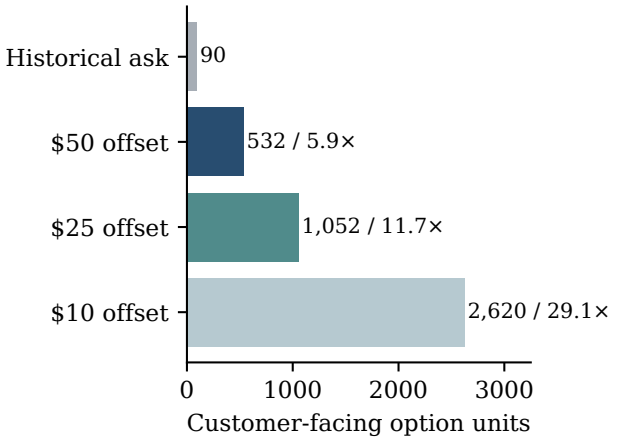


Figure 4: Conditional quotation capacity using approximately 60 ETH of initial hedging. The 90-unit historical ask is the comparison baseline; the three hedge scenarios assume executable offsetting calls. The $10 case is the aggressive scenario.

Quantities are rounded down to remain within approximately 60 ETH of initial hedging. The full measured execution cost is retained for these slightly smaller quantities.

The table describes calls offered in one customer-facing series, rather than a sum of gross orders across different books. However, each configuration also requires the corresponding quantity of hedge calls. The $25 configuration, for example, requires 1,052 executable hedge calls at the specified higher strike. Their availability and execution prices are assumptions, not observations established by the spot simulations.

Capital requirements for the combined position

To examine the financing mechanism, the portfolios are repriced under an illustrative stress grid covering spot changes between -20% and +20%, implied volatilities of

Table 4a. Hedge configurations and candidate quotation capacity

Hedge call strike above customer strike	Initial net delta per option package	Customer-facing call units supported	Total size relative to historical 90-unit ask	Initial spot execution cost per customer call
$50	0.1126	532	5.9×	$0.30
$25	0.0570	1,052	11.7×	$0.15
$10	0.0229	2,620	29.1×	$0.06

25%, 50%, and 75%, and remaining maturities of seven, three, one, and zero days. The initial spot hedge is held fixed within each scenario. The largest modeled losses are:

These calculations illustrate how a relatively small risk reserve can support a large customer-facing option position when an executable option hedge offsets much of its exposure. They are not production World margin outputs or maximum possible losses. Larger price shocks, volatility-surface distortions, liquidation costs, and hedge adjustments can require additional capital. In particular, the small initial delta difference need not persist as the underlying approaches either strike near expiry.

Risk capital must also be distinguished from cash financing. The spot purchase still needs funding. Assuming the option premiums settle together, the modeled net premium receipts reduce the required financing to approximately $139,000 in the $50 and $25 configurations, before dealer equity is applied to reduce borrowing. Joint margin can make this funding more capital-efficient, but does not eliminate the need for an executable financing facility.

Can the larger quotations remain competitively priced?

The historical Derive V2 observation reports a $67.20 ask for 90 units and a $63.30 bid for ETH-20261002-2700-C. [16] Its midpoint implies a $1.95 ask-side allowance if the midpoint is adopted as a hypothetical pre-cost option value. Here, that dollar allowance is used only as a pricing sensitivity: the historical contract and date differ from the new model portfolio.

At an assumed 5% annual borrowing rate for seven days, the measured spot friction and modeled financing expense consume the following amounts:

The hedge option's model premium is included in the portfolio cash flows. The remaining allowance must cover its execution cost above model value, subsequent hedging, venue fees, capital compensation, and profit.

The $25 configuration consequently identifies a concrete feasibility condition: a quotation of approximately 1,052 customer-facing calls can fit the illustrative allowance if the hedge option is available in that quantity and all remaining costs fit within approximately $1.67 per call. This is a materially larger opportunity than treating every customer call as a standalone short-convexity position requiring a full initial delta hedge.

From portfolio capacity to executable resting depth

A venue can expose these portfolios as resting option quotations by coordinating the customer order, option hedge, spot execution, and financing. Each quotation must remain backed by executable constituent liquidity and sufficient capital after adverse fills. Shared hedge orders cannot be counted repeatedly across different customer-facing series. When a constituent changes, the associated quotation must be updated, cancelled, or supported by committed dealer inventory.

The capacity of the route is therefore constrained by the smallest of its available hedge-option quantity, affordable spot execution, financing capacity, and residual-risk capacity. The 6–12-fold scenarios become executable depth only when all of these conditions hold together.

Derive V2 already recognizes option spreads and portfolio offsets. [11] The incremental World hypothesis is therefore that integrating constituent execution and financing makes more of this economically available capacity accessible as competitive resting orders. Establishing that advantage requires a matched comparison of the same contracts, prices, portfolios, and capital budgets, including Derive's deeper book and executable RFQs.

The numerical opportunity is nevertheless substantial: the measured spot route, combined with specified option offsets, supports candidate quotations of approximately 530–1,050 units with modest modeled residual capital requirements. The decisive empirical input is the executable supply of the offsetting options. The article's central proposition is that a venue can organize that supply, spot liquidity, and financing into a deeper options book through joint execution and portfolio margin.

5.4 Capital capacity and a matched dealer comparison

The Section 4 balance sheet supplies an independent test of the collateral constraint. Given $1.45 million in dealer capital, $1 million already in term loans, 98% term-claim eligibility in the integrated scenario, and an assumed

Table 4b. Illustrative joint stress reserves

Hedge strike difference	Modeled stress reserve per customer call	Reserve as a share of customer option reference notional	Total reserve for the candidate quotation
\$50	\$35.59	1.42%	\$18,932
\$25	\$17.49	0.70%	\$18,395
\$10	\$6.88	0.27%	\$18,034

Table 4c. Per-call pricing allowance

Hedge strike difference	Initial spot friction per call	Financing expense per call	Remaining from a **\$1.95** pricing allowance
\$50	\$0.30	\$0.25	\$1.40
\$25	\$0.15	\$0.13	\$1.67
\$10	\$0.06	\$0.05	\$1.84

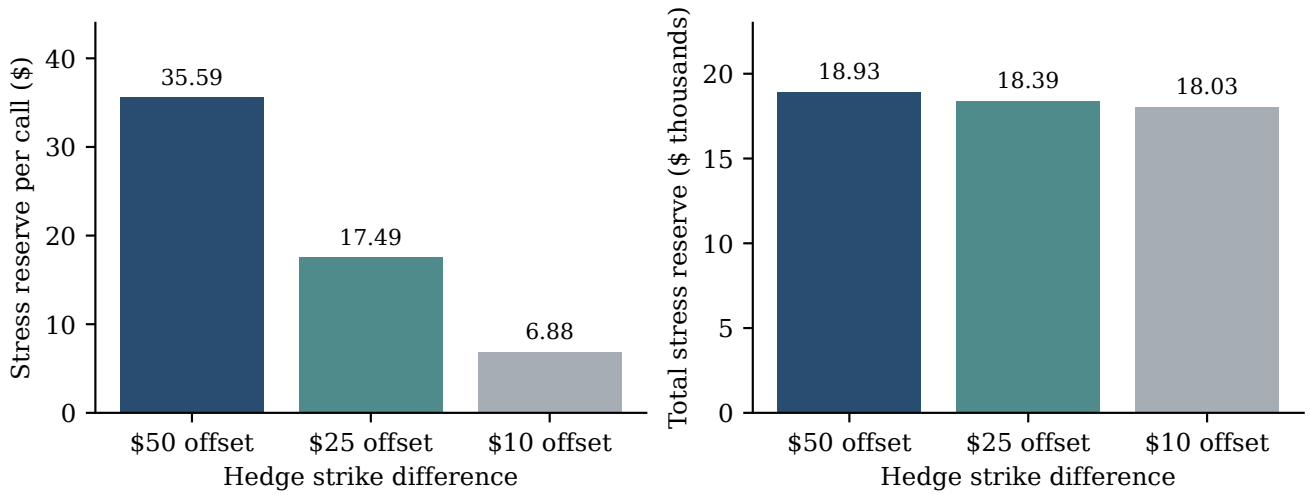


Figure 5: Modeled reserves from Section 5.3. Per-call risk falls as the strike gap narrows; aggregate reserves stay near \$19,000 as quoted units rise. These are illustrative stress outputs.

15% reserve per dollar of options reference notional, the capital ceilings are **\$3.00M** for segregated term credit and **\$9.53M** for eligible integrated credit. The ratio is **3.18x**, holding derivatives netting identical.

Now give both dealers the same accessible underlying hedge $H = \$3.00M$ and option $|\Delta| = 0.5$. Directional hedge capacity is $3/0.5 = \$6.00M$. With nonbinding residual-risk limits,

$$\begin{aligned} Q_{\text{separate}} &= \min(3.00, 6.00) = \$3.00\text{M}, \\ Q_{\text{integrated}} &= \min(9.53, 6.00) = \$6.00\text{M}. \end{aligned} \tag{15}$$

The integrated structure could therefore support **twice** the reference notional in this capital-bound scenario. If hedge liquidity fell to $H = \$1M$, both capacities would be limited to $2M$ and the uplift would vanish. This exercise isolates a fixed-term-claim eligibility mechanism **relative to a Derive-style derivatives account with that claim held externally**.

5.5 Immediate execution and incremental volume

Let X be an incoming sell order's option **reference notional**, and Q the firm bid quantity at a specified best price. Its immediate best-price fill is $\min(X, Q)$. For an **assumed** exponential order-size distribution with mean μ, the expected share of requested notional filled immediately is

$$F(Q, \mu) = \frac{E\left[\min(X, Q)\right]}{E[X]} = 1 - e^{-Q/\mu}. \tag{16}$$

Use $\mu = \$100,000$, round the historical bid to $Q_D = \$224,000$, and posit a new firm bid of $Q_U = \$1,000,000$ **at the same effective execution price**, replenished between requests. The resulting fill shares are 89.354% and 99.995%. Per $V = \$1B$ of hypothetical incoming sell demand,

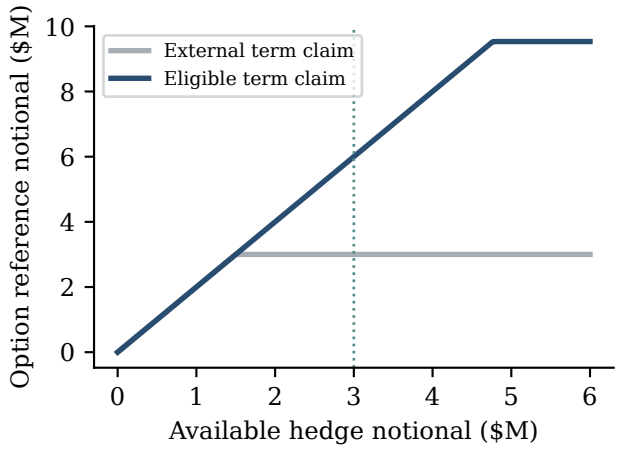


Figure 6: Capital and hedge constraints jointly determine dealer capacity. Curves use the Section 5.4 assumptions: delta 0.5 and capital ceilings of $3.00M and $9.53M.

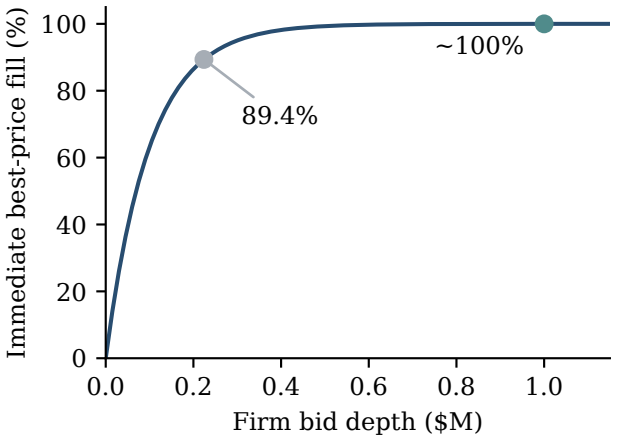


Figure 7: Modeled immediate best-price fill share, $1-e^{-D/\mu}$, with mean request size $\mu = \$0.10$M. Points mark $0.224M and $1M firm bid depth; this is a demand model.

$$\Delta V_{immediate} = V\ [F\,(Q_U, \mu) - F\,(Q_D, \mu)] = \$106.4M. \tag{17}$$

Some additional best-price execution would otherwise have occurred deeper in the book, via RFQ, or later. Define θ as the **assumed fraction becoming genuinely new volume**, not merely better-priced or accelerated execution. At $\theta = 50\%$,

$$\Delta V_{new} = \theta\, \Delta V_{immediate} = \$53.2M. \tag{18}$$

The separate $3M$-versus-$6M$ collateral experiment, with assumed mean request size $\mu = \$2M$, yields fill shares of 77.69% and 95.02%, or **$173.3M** more immediate best-price execution per $1B demand. At the same assumed conversion factor, the implied new trading is **$86.7M**. Neither calculation measures actual World routing or Derive order-flow conversion; both require fixed executable quote sizes and price conditions that are not provided by the aggregate dashboards.

5.6 Results and evidentiary status

6 The Economics of Financial Intermediation

The benefits of financing related derivatives exposures together are already material in traditional markets. Studies of clearing fragmentation and established cross-margin arrangements show how separate intermediation and collateral constraints can consume capital even where economic risks offset. The relevant gain is not the disappearance of risk, but the reduction of duplicated financing and operational requirements [5-7].

The BIS OTC figures imply $634B+$245.8B=$879.8B of average daily FX and interest-rate options notional, or approximately **$220T annualized** at a purely illustrative 250 trading days [8,9]. These are different statistical samples and measures: margin **savings** are collateral reductions, not fee revenue, while options **turnover** is neither financing cost nor profit.

A replication-aware venue potentially internalizes more than clearing efficiency: it can compare economically equivalent execution paths, source the associated spot and financing, and settle the combined package within one risk framework. Doing this across institutional FX or equity markets would also require matching legal, regulatory, market-access, and settlement rights; the present onchain estimates do not quantify those extensions.

7 World Markets V2: +Options

World Markets V2 applies these principles **at the venue level**. It will compare native option orders with executable synthetic constructions and assemble the latter from spot, fixed-term financing, and option payoffs whenever they provide a better price or deeper liquidity. Matching static-payoff components can be encoded in smart contracts. Where exact construction is unavailable, risk-bearing liquidity providers or vaults can hedge residual exposure through the existing spot and perpetual markets with full portfolio margin across all instruments.

Net exposures and eligible yield-bearing collateral can therefore support the combined trade rather than fund each leg independently; the existing risk framework recognizes eligible lender-side loans at 98% of value [20-22]. The expected result is greater executable depth and more efficient capital use, subject to sound options and credit-risk controls. The capital efficiency gains are illustrated in the table below, comparing the additional margin requirements on World Markets v2 vs Derive and isolated margin.

Table 5. Principal quantitative conclusions

Result	Estimate	Evidence
Onchain spot/options 30-day turnover	50.5x	Dated aggregate market data
Aave + Morpho loans / Derive options OI	10.8x	Dated outstanding balances
Static synthetic call bid/ask in worked example	\$3.50 / \$4.40	Illustrative executable-leg prices
Jointly margined candidate quote capacity	532–1,052 units	Modeled offsetting calls and approximately 60 ETH initial spot hedge
Ratio to historical Derive V2 90-unit ask	5.9–11.7x	Conditional capacity / historical displayed ask
Capital-limited capacity increase	3.18x	Fixed-term collateral eligibility scenario
Matched-case hedge-and-capital quote increase	2.00x	\$3M versus \$6M, conditional
Additional immediate best-price execution	\$106.4M / \$1B demand	Assumed order sizes and firm quote depth
Potential incremental traded volume	\$53.2M / \$1B demand	Further assumed 50% conversion

Table 6. Selected traditional-market scale and intermediation evidence

Measure	Reported observation
Clearing-fragmentation cost in studied LCH interest-rate swap market	\$80M per day
OCC-CME cross-margin average collateral savings (2014)	\$700M per day
OCC-CME average participant margin reduction	54%
CME swaps/futures average daily margin savings (2025)	\$8.4B
OTC FX options daily turnover (April 2025)	\$634B
OTC interest-rate options daily turnover (April 2025)	\$245.8B
Listed options contracts worldwide (2025)	88.65B contracts

Table 7. Capital required for complete trades (USD thousands; provisional comparison)

Trade	Position	Isolated	Derive	World Markets V2
Basis trade	\$100k spot + short perp	130	8.5	0.1
Covered call	\$100k spot + short ATM call	100	19	7
Collar	\$100k spot + long put + short call	100	13	4
Delta-hedged call	\$50k spot + short \$100k ATM call	80	9	2
Cash-secured put	\$100k term claim + short ATM put	130	12	9

8 Implementation, Risks, and Empirical Tests

A venue-level synthetic router must enforce **payoff identity, executable prices, and deliverable settlement**. Static legs must match currency, strike, expiry, exercise rights, and underlying distributions; borrowing or lending must provide the required fixed-date cash flows. Atomic or otherwise guaranteed bundled execution prevents one-sided fills. Firm synthetic depth must exclude the same liquidity reserved by another route, and funding markets must be executable for the full trade size.

For dynamically hedged options, a designated dealer must bear the gamma, volatility, gap, basis, funding, and liquidation risks. An account-level margin ledger must value options nonlinearly, recognize spot/perp offsets without ignoring basis risk, and stress loan claims under the same adverse states as the derivatives. Cross-product composability without shared enforceable margin rights does not solve this problem [19-22].

The mechanism admits direct tests: (i) compare each native option quote against contemporaneous executable synthetic route prices and *deduplicated* depth; (ii) measure dealer quotes and market impact conditional on available spot/perp hedge liquidity; (iii) stress the combined derivatives and credit account; and (iv) measure immediate fills and the fraction of improved execution that actually becomes new volume. A like-for-like performance comparison with Derive V3 must sample its firm displayed orders and RFQs at the same times and option contracts. If constituent-market execution is expensive,

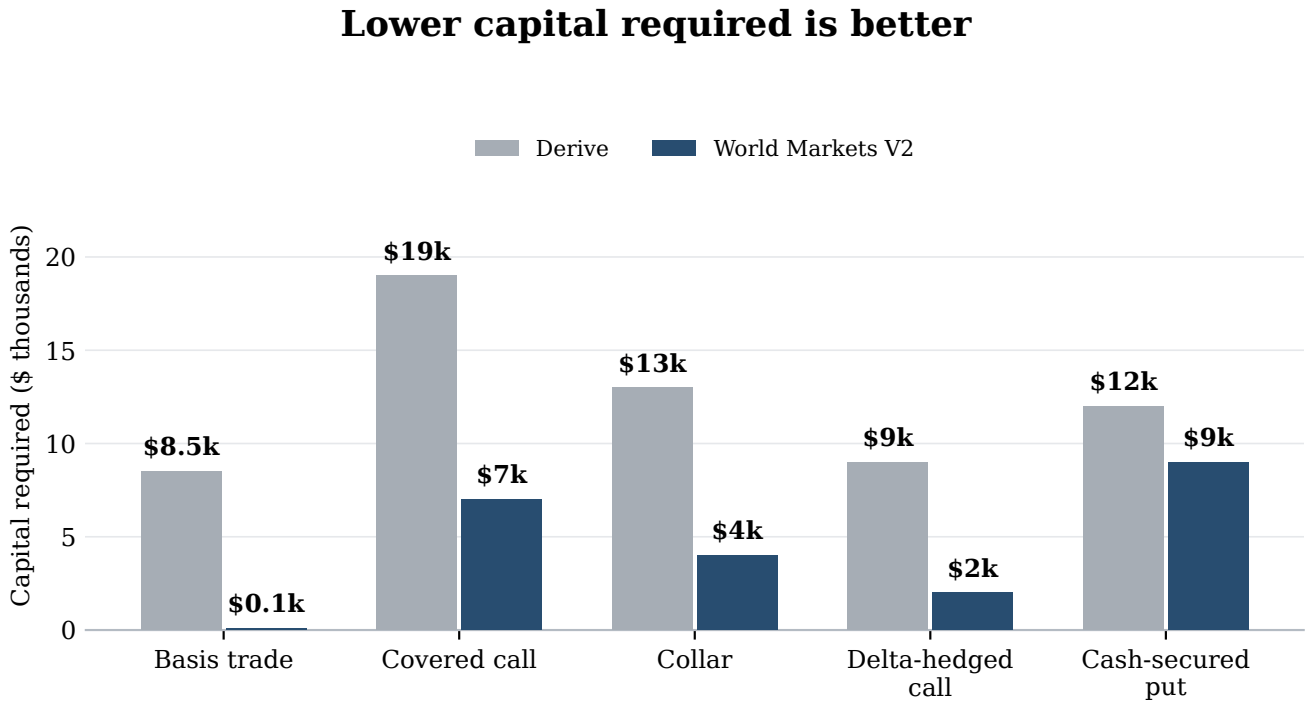


Figure 8: Capital required: Derive versus World Markets V2. Lower bars indicate less capital required for the same trade. The isolated-margin series is omitted for clarity; all values remain in Table 7. Values are provisional supplied estimates, not independently measured margin-engine outputs.

the financing is mismatched, or residual risk binds, the proposed liquidity advantage can disappear.

9 Conclusion

Put-call parity establishes that options, spot, and fixed-term repayment claims are linked by exact payoff identities. With an opposite option, or a compatible non-recourse contingent claim, a venue can construct an option payoff statically. Ordinary recourse loans supply the financing component but not convexity. When no exact static route exists, dynamically hedged dealers can supply further option liquidity while bearing the remaining nonlinear risk.

The market-structure implication is that **an options venue can source and execute liquidity from financially equivalent combinations of its other markets**, rather than rely only on orders native to each options series. Common portfolio margin can make these routes and their hedges less capital-intensive, but does not remove credit, volatility, or execution risk.

The paper's dated onchain data show a substantial existing base of spot and credit activity; its jointly margined scenarios illustrate 532–1,052 customer-facing option units (approximately 6–12 times the historical Derive V2 90-unit ask), supported by approximately 60 ETH of initial spot hedging and $19,000 in modeled stress reserves, a 3.18-fold capital-limited capacity increase under a specified term-credit example, and an estimated $53.2 million of new volume per $1 billion of incoming demand **only** with a 50% conversion assumption. The general contribution is a method for combining the existing financial system's constituent liquidity, rather than recreating it independently for every contingent contract.

References

[1] Stoll, H. R. (1969). "The Relationship Between Put and Call Option Prices." *Journal of Finance*. https://doi.org/10.1111/j.1540-6261.1969.tb01694.x

[2] Black, F., and Scholes, M. (1973). "The Pricing of Options and Corporate Liabilities." *Journal of Political Economy*. https://doi.org/10.1086/260062

[3] Merton, R. C. (1973). "Theory of Rational Option Pricing." *Bell Journal of Economics and Management Science*.

[4] Merton, R. C. (1974). "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." *Journal of Finance*. https://doi.org/10.1111/j.1540-6261.1974.tb03058.x

[5] Benos, E., Huang, W., Menkveld, A. J., and Vasios, M. (2024). "The Cost of Clearing Fragmentation." *Management Science*. https://doi.org/10.1287/mnsc.2023.4867

[6] Options Clearing Corporation (2015). "Enhancing the Resiliency of OCC and the U.S. Options Industry." https://www.theocc.com/newsroom/views/2015/05-07-enhancing-the-resiliency-of-occ-and-the-u-s

[7] CME Group (2026). "Portfolio Margining Onboarding:

Frequently Asked Questions." https://www.cmegroup.com/articles/faqs/portfolio-margining-onboarding.html

[8] Bank for International Settlements (2025). "Global FX Markets When Hedging Takes Centre Stage." https://www.bis.org/publications/qr-202512/global-fx-markets-when-hedging-takes-centre-stage

[9] Bank for International Settlements (2025). "OTC Interest Rate Derivatives Turnover in April 2025." https://www.bis.org/publications/202509-commentary-otc-derivatives-0

[10] Futures Industry Association (2026). "ETD Volume: December 2025." https://www.fia.org/fia/articles/etd-volume-december-2025

[11] Derive. "Introducing Cross-Asset Collateral and Portfolio Margin." https://blog.derive.xyz/introducing-cross-asset-collateral-portfolio-margin

[12] Derive. "Variable Rate Lending." https://insights.derive.xyz/variable-rate-lending/

[13] DefiLlama (snapshot cited: October 12, 2026). "Derive: Options Volume and Open Interest." https://defillama.com/protocol/derive

[14] DefiLlama (snapshot cited: October 12, 2026). "Options Notional Volume." https://defillama.com/options/notional-volume

[15] DefiLlama (snapshot cited: October 12, 2026). "Decentralized Spot Trading Volume." https://defillama.com/dexs

[16] GeorgL0ngGamma (September 24, 2026). "Paradex Execution Parity Adapter: Public Paradex/Derive Liquidity Snapshot." https://gist.github.com/GeorgL0ngGamma/4e00c477cb66424b9458827f8b3bec44

[17] DefiLlama (snapshot cited: October 12, 2026). "Morpho Active Loans." https://defillama.com/protocol/morpho

[18] DefiLlama (snapshot cited: October 12, 2026). "Aave V3 Active Loans." https://defillama.com/protocol/aave-v3

[19] Aave. "Health Factor and Liquidations." https://aave.com/help/borrowing/liquidations

[20] World Markets. "ATLAS: World's Risk Engine." https://docs.world.inc/tl-dr-unique-features/atlas

[21] World Markets. "Trading: Lending and Spot Mechanics." https://docs.world.inc/essentials/trading

[22] World Markets. "Market Structure: Fixed-Term, Fixed-Rate Lending." https://docs.world.inc/essentials/market-structure

[23] Options Industry Council. "Put/Call Parity." https://www.optionseducation.org/advancedconcepts/put-call-parity

[24] Options Industry Council. "Bear Call Spread (Credit Call Spread)." https://www.optionseducation.org/strategies/all-strategies/bear-call-spread-credit-call-spread